AutoResearch: Claim-Grounded Human–Agent Research Operations

Draft synchronized from node-local manuscripts

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Abstract

This draft evaluates AutoResearch at the level of schedulable research nodes, where each node carries local prompts, artifacts, review status, and closure conditions. The current TeX draft synchronizes the available node-local manuscripts into an exportable structure and preserves the evidence boundary: the only positive quantitative signal currently carried into the draft is a preliminary synthetic/offline single-run figure package, not a formal real-data result. The draft's process claim is that manuscript statements should remain traceable to evidence artifacts, review gates, response coverage, and limitation records before stronger claims are allowed.

1 Introduction

AutoResearch addresses a practical bottleneck in agent-assisted research: a fluent manuscript can hide whether claims, evidence, protocols, reviews, and responses are actually connected. Reporting and peer-review policies already expect inspectable claims, reproducible evidence, and transparent limitations [1, 2]. The system therefore treats a research node as the unit of work. Each node has a path, prompt assets, local artifacts, review status, and explicit closure conditions. This design makes a paper draft inspectable as a chain of bounded research decisions rather than a single prose surface.

The draft makes a narrower process contribution: it specifies that substantive manuscript claims should point to evidence, evidence items should retain explicit boundaries, and phase transitions should be backed by review evidence or recorded gate logic. In the current repository state, that contribution is supported as a process and documentation claim. It is not yet supported as a final real-data performance claim.

The current TeX synchronization carries forward only the evidence already accepted by upstream nodes. Earlier P1 nodes established data provenance, pseudocode, repository blueprinting, bounded synthetic/offline validation, result-state separation, and a draft result figure. The only quantitative signal allowed into this draft is the P1_09 figure package derived from P1_04/P1_05 synthetic/offline evidence. The introduction therefore frames the paper around auditability and research-operation discipline, while reserving stronger empirical claims for future accepted formal rows.

2 Preliminaries

A research node is a schedulable unit with a local manuscript, acceptance checklist, review rubric, artifacts, and status. A node may reach author exit when its local outputs are coherent, but node closure requires review evidence and response coverage. This distinction prevents an authoring agent from treating generated text as accepted research truth.

A claim-evidence registry records claim identifiers, claim types, support status, manuscript location, actions, and evidence references. A figure manifest records figure identifiers, claim references, evidence references, source paths, output paths, and quality checks. These registries are part of the scientific record for this workflow: they define what a paragraph or figure is allowed to say.

Evidence status is intentionally stricter than narrative plausibility. Supported evidence can be used only within its documented boundary. Unsupported and uncertain evidence remain visible as limitations or deferred work. For example, the P1_09 result figure is a limited proxy for a synthetic/offline single-run signal. It is not evidence for real-data generalization, RM101 resolution, selected-backend readiness, or final submission fitness.

3 Methods

The study design evaluates AutoResearch as a documented human–agent research workflow rather than as a model-training recipe. The unit of analysis is the research node. A node is eligible for evaluation only when it has a node path, research prompt, acceptance checklist, allowed edit scope, expected outputs, evidence artifacts, and reviewer or human-gate status. This prevents generic prose from being scored without a verifiable research state and follows the local node-review rubric used by this repository [3].

The intervention is the documented AutoResearch node package, which includes local framing, claim/evidence/protocol registries, pre-execution gates, independent review, and response coverage; it is not a test of a language model alone. Comparison conditions should include manual checklist workflow, prompt-only agent workflow, and agent workflow without an independent gate, using the same source prompt, checklist, rubric, budget, artifact audit procedure, and output contract.

Author exit and node close are separated. Author exit means that local artifacts are present and internally coherent. Node close requires reviewer verdict or human-gate evidence. This distinction is a method variable because premature phase transition is one of the risks under study.

The primary outcome is claim-evidence validity rate: the proportion of eligible claims that have a claim identifier, evidence identifier, artifact path, citation status when applicable, boundary, and reviewer or human-gate status. Secondary outcomes include protocol completeness, independent reviewer pass rate, reproducibility rerun agreement, unsupported claim count, hard-fail closure rate, human-gate escalation rate, and time to author exit.

The procedure is fixed before result interpretation. First, freeze the node set, prompts, checklists, rubrics, budget, reviewer assignment rule, and artifact paths. Second, run each condition on the same node set and collect node-local artifacts, claim/evidence/protocol links, gate outcomes, and reviewer decisions. Third, audit each eligible claim against its evidence location and boundary. Fourth, report point estimates, uncertainty when available, missing-data policy, and negative results. Fifth, map reviewer concerns to claim identifiers, evidence identifiers, manuscript locations, actions, response status, and closure evidence.

Reproducibility requires repository version, node paths, fixed node set, prompts, acceptance checklists, review rubrics, allowed edit scope, budgets, random seeds or reviewer assignments, exact run commands or manual ledgers, model/provider metadata when agents are used, artifact collection rules, evaluator versions, and result ledger schema. Graph, Canvas, and dashboards are projections or operation surfaces; they are not sources of research truth.

4 Results

The current result content is limited to a preliminary synthetic/offline signal carried forward from the P1_04 and P1_05 evidence chain and visualized in P1_09. Figure 1 is the first TeX callout for this signal. The source ledger contains one baseline row, simple_fullchain on Ottawa synthetic data with offline stub mode, and one controlled attempt row, supervisor_proving under

Workflow row	Test accuracy	Test macro-F1
simple_fullchain baseline	0.8333	0.8286
supervisor_proving attempt	1.0000	1.0000

Figure 1: Preliminary synthetic/offline signal carried into the TeX draft. claim_ref: c1. evidence_ref: P1_09_E004_FIGURE_DATA_TSV. Source/provenance: P1_04 auto-experiment ledger, P1_05 result registry, and P1_09 figure manifest. The figure reports no variance, no real-data row, no RM101 resolution, and no selected-backend decision.

the same synthetic/offline boundary. The controlled attempt is higher on the recorded accuracy and macro-F1 values, but this is a single-run signal rather than a formal result.

The P1_09 figure package maps this signal to upstream claim c1 only as a supported limited proxy. It also records the missing evidence: no accepted real-data formal row, no repeated-run variance, no RM101 resolution, no selected-backend lock, and ambiguity around the perfect synthetic score. The TeX draft therefore reports the signal as a candidate reason to continue the mechanism path, not as proof of general performance improvement.

The discussion reports both the bounded synthetic/offline signal and the retained unsupported or uncertain evidence; unsupported rows are not rewritten as support. This is central to the AutoResearch argument. A submission-quality empirical claim will require accepted Stage C main-result rows, Stage D ablation rows, adapter preflight closure, and review scores that meet the final submission threshold.

5 Discussion

The synchronized draft separates process claims from performance claims. It can argue that node-local contracts, evidence registries, and independent review gates make the research process inspectable. It cannot yet argue that AutoResearch improves real-data model performance, resolves RM101 evidence, locks a selected backend, or passes formal Stage C/D evaluation. Those limitations remain explicit so that negative and uncertain evidence stays part of the manuscript record rather than being converted into stronger claims.

Data availability

The data for this draft are repository-local research artifacts, node prompts, checklists, review verdicts, response files, result ledgers, figure data files, and TeX synchronization maps. The current draft references the P1_04 synthetic/offline result ledger and the P1_09 figure package as local evidence artifacts. No external private data are redistributed by this TeX node.

Code availability

The code relevant to this TeX synchronization node is the repository validation and export metadata, including the P2_03 section and sync maps, the P1_09 deterministic figure renderer, and the submission validation script. Graph, Canvas, web dashboard, and other projections are not treated as sources of research truth.

References

[1] Nature Portfolio. Reporting standards. `https://www.nature.com/nature-portfolio/editorial-policies/reporting-standards`.

[2] IEEE Author Center. Submission and peer-review policies. `https://journals.ieeeauthorcenter.ieee.org/become-an-ieee-journal-author/publishing-ethics/guidelines-and-policies/submission-and-peer-review-policies/`.

[3] Local repository evidence gate. `test/NATURE_LEVEL_NODE_RUBRIC.md` and node-local review rubrics.